

# SHAALE-VIKAS

Android App Development using GenAI

B.E. Final Year Project | Project Title: 02

## 1. PROJECT OVERVIEW

**Shaale-Vikas** is a School-Alumni Bridge platform designed as an Android application that leverages GenAI (Generative Artificial Intelligence) to create transparency between rural school administrators and alumni networks. The application addresses the critical gap between school infrastructure needs and timely funding by enabling real-time visibility into micro-repair needs, alumni pledge tracking, and impact documentation through before/after photography.

## 2. PROBLEM STATEMENT

Rural schools face urgent infrastructure and repair needs (e.g., leaking roofs, broken desks, damaged furniture) that significantly impact the learning environment. While government funding mechanisms exist, the approval and disbursement process is slow, often taking months or years. Meanwhile, alumni of these schools—now working professionals—want to contribute and support their alma maters but lack clear visibility into what the school actually needs. This disconnect results in:

- School infrastructure issues remaining unaddressed for extended periods
- Alumni networks underutilized and disconnected from their schools
- Lack of transparency about school needs and fund utilization
- No mechanism to track and verify the impact of alumni contributions

## 3. DETAILED VISION & SOLUTION

**Shaale-Vikas** (which means 'School Development' in Hindi) is a transparency and trust-building platform that acts as a bridge between school headmasters and alumni. The core vision is to empower alumni networks to become active, engaged support systems for their schools through:

### For Headmasters (Admin Role):

Create and maintain a dynamic 'Needs Dashboard' listing micro-repairs or improvements required at the school. Each need includes a description, cost estimate (with AI-assisted suggestions), and current collection progress. Headmasters can easily update this list and upload before/after photos to verify completion of work.

### For Alumni (User Role):

Browse the needs dashboard, see real-time progress towards each goal (e.g., '60% funds collected for Toilet Repair'), and pledge support by clicking a button. Alumni appear in a public 'Hall of Fame' list, creating social recognition and community cohesion. The pledge mechanism is simulated (no real

money exchange) to keep the project scope manageable.

## 4. KEY APPLICATION FEATURES

- **Needs Dashboard (Admin):** Headmaster can post, edit, and manage school repair/improvement needs with cost estimates. Interface is simple and admin-friendly.
- **Real-time Progress Bar:** Each need displays the percentage of funds pledged. Firebase Realtime Database ensures live updates across all user devices instantly.
- **Pledge System:** Alumni click 'Pledge' button to contribute (simulated, no payment). Pledge is recorded in Firebase and instantly reflected in progress calculations.
- **Donor Hall of Fame:** A public list of all alumni who have pledged, sorted by contribution amount or date. Builds community recognition and motivates further contributions.
- **Impact Photos (Before/After):** Headmaster uploads before photos when posting a need and after photos when work is completed. Provides visual proof of impact and builds donor trust.
- **Role-Based Authentication:** Two roles via Firebase Auth: Admin (Headmaster) with full CRUD access to needs, and Alumni (Donors) with read+pledge permissions.

## 5. USER FLOWS

### Headmaster (Admin) Flow:

Login (email/password or Google Sign-In) → Post Micro-Need with Title, Description, Cost Estimate → Upload Need Photo (from camera/gallery) → Monitor Pledges in Real-time → Upload After Photos → View Completion Status

### Alumni (Donor) Flow:

Login → View Needs Dashboard → Explore School Needs with Cost & Progress → View Before Photos → Click 'Pledge' Button → Appear in Hall of Fame → View After Photos (completed work)

## 6. COMPLETE SOFTWARE & TECHNOLOGY STACK

| Technology                             | Role & Details                                                                                                                    |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Android Studio (Kotlin/Java)           | Primary IDE for building the Android application. Kotlin is recommended for modern Android development.                           |
| Firebase Realtime Database / Firestore | Cloud NoSQL database storing all app data: school needs, pledge records, donor information, etc.                                  |
| Firebase Authentication                | Secure user authentication with two roles: Admin (Headmaster) and Alumni (Donors). Supports email/password and social logins.     |
| Firebase Cloud Storage                 | Stores all uploaded images (before/after photos). Integrated with Android ImagePicker for seamless uploads.                       |
| Google Gemini API (GenAI)              | Core AI component. Assists headmasters by: auto-generating cost estimates based on need descriptions.                             |
| Retrofit + OkHttp                      | HTTP client library for making REST calls to Gemini API from Android. Handles async networking.                                   |
| Material Design 3                      | Google's official Android UI framework providing polished components (Progress Bars, CardView, etc.).                             |
| ImagePicker Library                    | Android library enabling users to select or capture images from device camera or gallery. Integrated with Firebase Cloud Storage. |

### Supporting Tools:

- **Figma:** Design wireframes and high-fidelity mockups before development. Essential for project documentation and viva presentation. Shows professional UI/UX planning.
- **Git + GitHub:** Version control for code collaboration, code reviews, and project submission. Demonstrates professional development practices.
- **Android Emulator / Physical Device:** Testing and debugging the application during development.

## 7. IMPACT GOALS & SUCCESS CRITERIA

### Community Ownership:

Empower citizens and alumni to take active ownership of school development rather than waiting for government funds.

### Education Quality:

Ensure that infrastructure gaps don't negatively impact student learning outcomes. Quick turnaround on repairs = better learning environment.

### Transparency & Trust:

Build trust through visual proof (before/after photos). Donors see exactly where their contributions go and what impact they achieve.

## 8. SUCCESS CRITERIA FOR BE FINAL YEAR PROJECT

- **Needs Dashboard must be easily editable by the Admin** – Headmaster can add, update, delete needs without coding knowledge. Uses Android UI patterns (FAB, dialogs, forms).
- **Pledge mechanism must not involve real money** – Simulated donation system. Pledge is a boolean flag in Firebase, no payment gateway integration required.
- **UI must be professional and inspiring** – Use Material Design 3 components, consistent typography, appropriate color scheme, and smooth animations. Should feel like a real-world app.
- **Real-time updates** – Progress bars and donor lists update instantly via Firebase Realtime DB. No need to refresh manually.
- **GenAI Integration** – At least one meaningful AI feature (e.g., cost estimate suggestion, photo analysis). Demonstrates AI/ML application knowledge.
- **Role-based access control** – Different UI/permissions for Admin vs. Alumni roles. Demonstrates security and access control understanding.

## 9. IMPLEMENTATION RECOMMENDATIONS

**Phase 1 – Setup & Backend:** Set up Firebase project, design Firestore schema, configure authentication. Create Android project structure with Kotlin, add Firebase SDKs.

**Phase 2 – Core UI & Authentication:** Build login/signup screens, role-based UI navigation. Implement Admin and Alumni dashboards.

**Phase 3 – Feature Implementation:** Build Needs Dashboard (CRUD), Progress Bar (real-time), Pledge System, Hall of Fame list, Photo Upload (with Firebase Storage).

**Phase 4 – AI Integration:** Integrate Gemini API for cost estimation and photo analysis. Test and optimize.

**Phase 5 – Polish & Testing:** Refine UI/UX, test on multiple devices, prepare documentation, viva presentation, and project report.

## 10. CONCLUSION

Shaale-Vikas is a well-scoped, socially impactful Android project that combines modern technology (Firebase, GenAI, Material Design) with a meaningful real-world use case. The project demonstrates proficiency in mobile development, cloud backend integration, AI API consumption, and user-centered design—all essential skills for professional software engineers. The combination of Firebase for real-time backend and Gemini API for GenAI integration provides sufficient technical depth for a strong final year project viva while remaining achievable within typical project timelines.